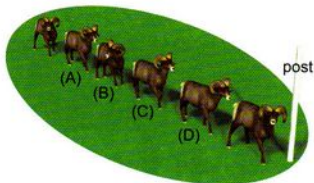


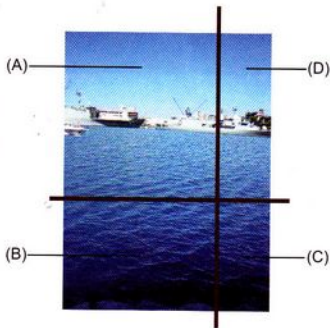
1. These sheep are walking in a line past a post.

Which will be the fourth sheep to walk past the post?



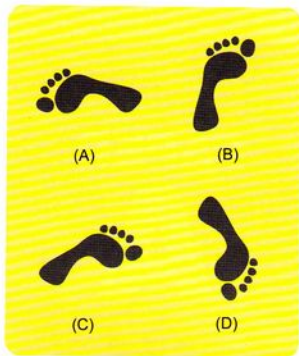
2. By cutting along the lines, this picture will be cut into four pieces.

Which piece will be the largest?



3. Three of these pictures show a footprint made by Lisa's left foot.

Which picture shows a footprint made by Lisa's right foot?



4. When a person has a birthday in Candle Land, the birthday cake is decorated with three times as many candles as the person's age.

Adam lives in Candle Land.

If Adam is turning nine, how many candles will he have on his birthday cake?

- (A) 3
(B) 9
(C) 18
(D) 27

5. Manling made this spinner.



Manling spins the spinner.

Which symbol does she have the least chance of landing on?

- (A) 
 (B) 
 (C) 
 (D) 

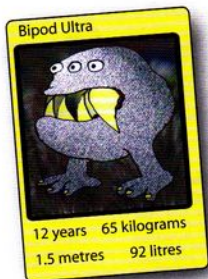
7. Adam made a pentagonal prism.



Which of the following shows all the faces of his prism?

- (A) 
 (B) 
 (C) 
 (D) 

8. Finn collects monster cards like this one.



What is the mass of the monster?

- (A) 12 years
 (B) 65 kilograms
 (C) 1.5 metres
 (D) 92 litres

6. Which of these does **NOT** show a half?



(A)



(B)



(C)



(D)

9. Luisa is planning to take one ball out of a container without looking.

For which container is the chance of taking out a white ball unlikely, but not impossible?



(A)



(B)



(C)



(D)

10. Who is sitting north-east of Luisa?



Adam



Noor



Luisa



Sunil



Manling

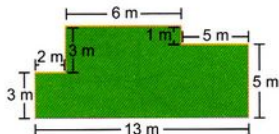
- (A) Adam
(B) Manling
(C) Noor
(D) Sunil

11. A bottle can hold 1 litre of juice. It is half full.

How many millilitres of juice are in the bottle?

- (A) 5
(B) 50
(C) 500
(D) 5000

12. Here is Noor's backyard.

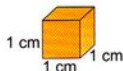


Noor wants to fence her backyard.

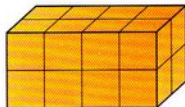
How many metres of fence does she need?

- (A) 27
(B) 31
(C) 35
(D) 38

13. This is a cube.

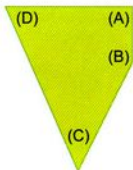


How many of these cubes make up this solid?



- (A) 12
(B) 16
(C) 20
(D) 48

14. Which angle inside this shape is a right angle?



15. Sunil had some apples. He shared them equally with his friend Manling.

Manling got 16 apples from Sunil.

How many apples did Sunil have at the beginning?

- (A) 8
(B) 16
(C) 22
(D) 32

16. How many complete circles have been drawn to make this pattern?



- (A) 6
(B) 7
(C) 12
(D) 13

17. How many lines of symmetry does this shape have?



- (A) 1
(B) 2
(C) 3
(D) 4

18. What is the next shape in this pattern?



(A)



(B)

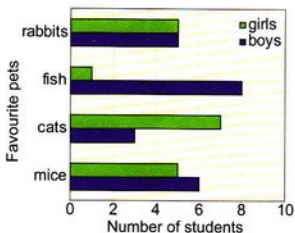


(C)



(D)

19. Some students were asked to name their favourite type of pet.



Which type of pet was named by the greatest number of students?

- (A) rabbits
(B) fish
(C) cats
(D) mice

21. Manling, Sunil and Adam ate some oranges that had been cut into quarters.



Manling ate five orange quarters.
Sunil ate six orange quarters.
Adam ate nine orange quarters.

How many whole oranges were eaten altogether?

- (A) 4
(B) 5
(C) 10
(D) 20

20. Bill has a dice with the numbers 1 to 6 marked on it.



Bill rolls the dice.

Which kind of number is the dice most likely to land on?

- (A) a number less than 3
(B) a number less than 4
(C) a number less than 5
(D) a number less than 6

22. Which clock **incorrectly** shows the time that is written under it?



(A)



(B)



(C)



(D)

23. Luisa knows that $40 \times 14 = 560$. Using this information, she worked out 39×14 .

What calculation could Luisa have used to get the correct answer?

- (A) $560 - 40$
- (B) $560 - 39$
- (C) $560 - 14$
- (D) $560 - 1$

24. Amy wrote down a number sentence, but one digit was rubbed out.

$$176 + 2\text{ }4 = 430$$

What is the missing digit?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

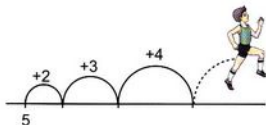
25. Noor has \$5 to buy lollipops. She wants to buy as many lollipops as she can with her money.

If each lollipop costs 80c, how much change will she have left?

- (A) 20c
- (B) 40c
- (C) \$1.00
- (D) \$4.20

26. Adam is making a pattern by jumping on a number line and writing down the number he lands on.

He starts on the number 5 and jumps as shown in the diagram.



On what number will Adam land on his fourth jump?

- (A) 19
 - (B) 18
 - (C) 17
 - (D) 14
27. Luisa's school starts at 9 am. Luisa arrived at school 80 minutes late.

What time did she arrive?

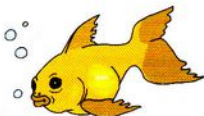
- (A) 9:20 am
- (B) 9:50 am
- (C) 10:20 am
- (D) 10:50 am

28. Adam and Sunil were playing a game. Adam scored 352 points while Sunil scored 268 points. They recorded the sum and the difference of their scores.

Which option shows what they recorded?

	Sum	Difference
(A)	620	84
(B)	620	116
(C)	510	84
(D)	510	116

29. At Pete's Pet Shop there is a rule that there can only be a maximum of 8 gold fish in a tank at any time.



Next month there will be 35 gold fish at Pete's Pet Shop.

How many tanks will Pete need?

- (A) 1
- (B) 4
- (C) 5
- (D) 8

30. Sunil had a 20 m roll of masking tape. He cut off 15 cm.

How much tape is left on the roll, in cm?

- (A) 1985
- (B) 500
- (C) 185
- (D) 5

31. A group of students planted trees around their school.



They planted 40 trees altogether. They planted one gum tree for every 4 wattle trees.

How many wattle trees were planted?

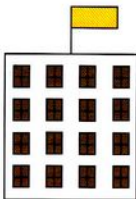
- (A) 36
- (B) 32
- (C) 10
- (D) 8

32. Adam put an apple on a set of scales. It had a mass of 316 grams. He then put a pear on the scales as well. Together the apple and pear had a mass of 613 grams.

How many **more** grams was the apple than the pear?

- (A) 13
(B) 19
(C) 297
(D) 303

33. This picture shows a building with 16 windows and a flag on top. The total area of the 16 windows is 32 square metres.



What is the area of the flag, in square metres?

- (A) 2
(B) 3
(C) 6
(D) 8

34. Adam is hiding a 3-digit number. Here are a few hints to find out the number he is hiding:

- all the digits are odd numbers
- it can be divided by 5 with no remainder
- the difference between the hundreds digit and the tens digit is the same as the difference between the tens digit and the ones digit.

Which could be Adam's number?

- (A) 395
(B) 630
(C) 951
(D) 975

35. Sunil has a mirror facing his wall clock.

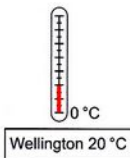
The picture shows the reflection of his clock in the mirror.



What is the actual time?

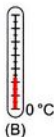
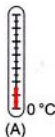
- (A) 9:50
(B) 9:10
(C) 8:50
(D) 3:10

36. This thermometer shows the temperature in Wellington. The numbers have been left off the scale.



The temperature in Brisbane is 4°C warmer than in Wellington.

Which thermometer shows the temperature in Brisbane?



37. Manling had some marbles. She gave Noor one-half of her marbles.

Then she gave Sunil one-third of the marbles that she had left.

After that Manling had 14 marbles.

How many marbles did Manling start with?

- (A) 10
(B) 18
(C) 42
(D) 84

38. Sunil and Luisa went to get a double scoop of ice-cream each.

Sunil bought his ice-cream from a shop that had three different flavours: blueberry, vanilla and chocolate. He could therefore choose from six different combinations.

KEY

B = Blueberry
V = Vanilla
C = Chocolate

BB, BV, BC
VV, VC
CC

Luisa went to another shop which had 10 different flavours.

From how many different combinations could she choose?

- (A) 100
(B) 55
(C) 45
(D) 20

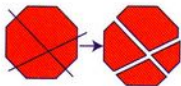
39. The picture shows views of a cube from three directions.



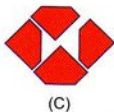
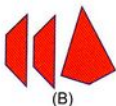
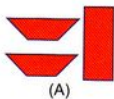
Which of these options shows the view of the cube from the west?



40. The diagram shows how an octagon can be cut with two straight cuts.



Using two different straight cuts, which of these sets of shapes **cannot** be made from an octagon?



NO	ICAS MATHS PAPER A - 2010 ANSWER SHEET
1	B
2	A
3	A
4	D
5	C
6	B
7	C
8	B
9	A
10	C
11	C
12	D
13	B
14	A
15	D
16	B
17	B
18	A
19	D
20	D
21	B
22	C
23	C
24	C
25	A
26	A
27	C
28	A
29	C
30	A
31	B
32	B
33	C
34	D
35	C
36	B
37	C
38	B
39	A
40	D